

METHODS

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1 SAMPLING AND FIELD PARAMETERS

[routine measurements only – adjust text to your actual work!]

Water sample were taken mid-stream wherever possible and filtered through 0.45 μm syringe disk filters (Minisart HighFlow PES, Sartorius AG, Germany). Samples for major ions and trace metal were filled into 15 mL polypropylene conical centrifuge tubes (VWR® Ultra-High Performance, HPDE-Cap with sealing ring, VWR International GmbH, Darmstadt, Germany). For oxygen and hydrogen stable isotopes of water ($\delta^2\text{H}_{\text{H}_2\text{O}}$ and $\delta^8\text{O}_{\text{H}_2\text{O}}$) samples were collected in 12 mL glass bottles. Additional samples were taken for the concentration and stable isotope analysis ($\delta^{13}\text{C}$) of dissolved inorganic (DIC) and organic carbon (DOC) in 40 mL amber borosilicate glass vials that fulfill the specifications of the US Environmental Protection Agency (so-called EPA vials). The open-hole caps of the EPA vials were equipped with flipped butyl rubber-PTFE septa, the butyl rubber side facing inside, to improve gas tightness. Beforehand, the EPA vials were poisoned with a few drops of nearly saturated HgCl_2 solution ($\sim 50 \text{ g L}^{-1}$), to prevent any microbial alteration of the carbon species during sample storage. EPA vials were filled to maximum capacity and capped without air bubbles. All of the above-described sample containers were tightly sealed with Parafilm® and stored cool ($\sim 4^\circ\text{C}$) and dark.

Samples for the analyses of $^{18}\text{O}/^{16}\text{O}$ ratios of dissolved oxygen ($\delta^{18}\text{O}_{\text{DO}}$) were collected in 12-mL Labco Exetainers™ (Labco Ltd. Lampeter, U.K.) that were pre-poisoned with a saturated HgCl_2 solution in the laboratory. The vials were completely filled with a positive meniscus by a syringe and immediately capped in field using screw caps with a butyl septum. Contamination by atmospheric O_2 was negligible. A comparison with previous measurements of submerged and post-poisoned samples did not show a significant change of the oxygen concentration nor the isotopic composition

Table 1. Sampling equipment used in 2023 field campaigns.

Parameter	Vial type	Material	Volume
Major ion	Centrifuge tubes	PP	15 mL
Trace elements	Centrifuge tubes	PP	15 mL
$\delta^8\text{O}$ and $\delta^8\text{H}$ water	Glass with butyl rubber septa and cap / Labco Exetainer	soda glass / borosilicate glass	10 mL / 12 mL
$\delta^{13}\text{C}$ -DIC and DOC	EPA-type vials with butyl rubber septa	borosilicate glass	40 mL
$\delta^8\text{O}$ -DO	Labco Extainer	borosilicate glass	12 mL

Measurements of temperature, pH, electrical conductivity (EC), and dissolved oxygen (DO), and reduction-oxidation potential (Eh) were conducted in the field by a multi parameter instrument ([*select your models:*] Hach HQ40d multi or HQ2200, Hach Company, Loveland, CO, USA or WTW Multi 3620 IDS or WTW 350i, Xylem Analytics Germany, Weilheim, Germany). All probes were calibrated prior to each field campaign. The pH and conductivity probes used built-in automatic temperature compensation functions, and temperature precision is quoted as $\pm 0.1^\circ\text{C}$ by the manufacturer. Conductivity values were referenced to 25°C by a non-linear correction function (nLF) for natural waters. Instrument precision ($\pm 1\sigma$) is about 0.05 for pH and 2 % for EC, DO and Eh.

Total alkalinity (TA) was determined from 100 mL water samples directly in the field by a Hach Digital Titrator (Model 1690001, Hach Company, Loveland, CO, USA). The titration endpoint was determined by color changes of a pH indicator (bromocresol green-methyl red). At a pH < 9, hydroxide (OH^-) does not contribute to TA at an appreciable concentration and other TA contributing species (BOH_4^- , $H_3SiO_4^-$) can be regarded as negligible in the waters examined by this study. Therefore, TA essentially contains only carbonic alkalinity and the titration value was used for the calculation of carbonate species concentrations, which is mainly bicarbonate (HCO_3^-) in the observed pH range (see Verma et al., 2015).

[needs to be included here: laboratory-based Eco Titrator (Deutsche Metrohm GmbH & Co. KG, Filderstadt, Germany)]

2 AQUATIC CHEMISTRY

Ion chromatography (IC)

Major anions (F^- , Br^- , Cl^- , NO_3^- , NO_2^- , SO_4^{2-} , PO_4^{3-}) were determined by ion chromatography (930 Compact IC Flex, Deutsche Metrohm GmbH & Co. KG, Filderstadt, Germany). Prior to ion chromatography analysis, samples were filtered with 0.45 μm disk filters, which were flushed with sample before the autosampler vials were filled. Limit of quantification (LOQ) for major ions was 0.1 mg L⁻¹ with a typical precision of <5% (1 σ) relative standard deviation (RSD) based on the repeated analysis of two control standards treated as unknowns in the lower and upper calibration range.

Inductive coupled plasma mass spectrometry (ICP-MS)

Major cations (Na^+ , K^+ , Li^+ , Ca^{2+} , Mg^{2+}) and trace elements were analyzed using an iCAP Qc inductively coupled plasma - mass spectrometer (ICP-MS) (Thermo Fisher Scientific Inc., Bremen, Germany) fed by a SC-2DXS autosampler (Elemental Scientifics, Omaha, NV, U.S.A). Samples were measured three times and stabilized by two drops of 65% HNO₃ Suprapur. The typical relative standard deviation (RSD) for three replicates was better than 1 %. For most elements the practical quantification limit is 0.1 $\mu g L^{-1}$. Operating conditions of the instrument are reported in Table 2.

Table 2. Operating conditions ICP-MS.

Model	iCAP Qc ICP-MS
Manufacturer	Thermo Scientific
Autosampler	ESI SC-2DXS
Plasma Power	1550 W
Cool gas flow	14 L min ⁻¹
Auxiliary gas flow	0.65 L min ⁻¹
Nebulizer gas flow	1.03 L min ⁻¹
CCT (KED mode) flow	5 L min ⁻¹
CCT gas	8 % H ₂ in He
Sampler/Skimmer	material: Nickel
Spray Chamber Temperature	2,7 °C
Dwell time	10 ms (40ms for Se, As)
Number of replicates	3
RSD (3 replicates) typical	< 1%
Number of Sweeps	70
Sample flow	0.4 ml min ⁻¹
LOD typical	< 0.1 ppb